

4.2 Forces: Density & Pressure

Question Paper

Course	CIEA Level Physics
Section	4. Forces, Density & Pressure
Topic	4.2 Forces: Density & Pressure
Difficulty	Medium

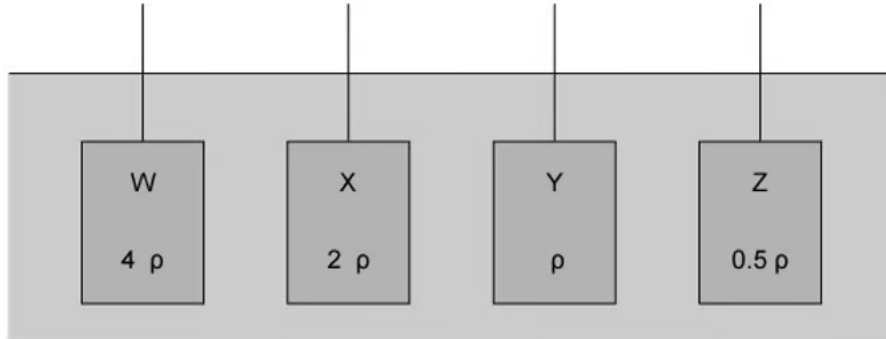
Time allowed: 20

Score: /10

Percentage: /100

Question 1

Four cuboids with identical length, breadth and height are immersed in water. The cuboids are held at the same depth and in identical orientations by vertical rods, as shown.



Water has density ρ

Cuboid W is made of material of density 4ρ

Cuboid X is made of material of density 2ρ

Cuboid Y is made of material of density ρ

Cuboid Z is made of material of density $\frac{1}{2}\rho$

Which statement is correct?

- A. the upthrust of the water on each of the cuboids is the same
- B. the upthrust of the water on W is twice the upthrust of the water on X
- C. the upthrust of the water on Z is twice the upthrust of the water on Y
- D. the upthrust of the water on Y is zero

[1 mark]

Question 2

Atmospheric pressure at sea level has a value of 100 kPa.

The density of sea water is 1020 kg m^{-3}

At which depth in the sea would the total pressure be 150 kPa ?

- A. 5 m
- B. 25 m
- C. 50 m
- D. 55 m

[1 mark]

Question 3

A beam of α -particles collides with a lead sheet. Each α -particle in the beam has a mass of $6.6 \times 10^{-27} \text{ kg}$ and a speed of $1.5 \times 10^7 \text{ m s}^{-1}$

5.0×10^4 α -particles per second collide with an area of 1.0 cm^2 of lead. Almost all of the α -particles are absorbed by the lead so that they have zero speed after collision.

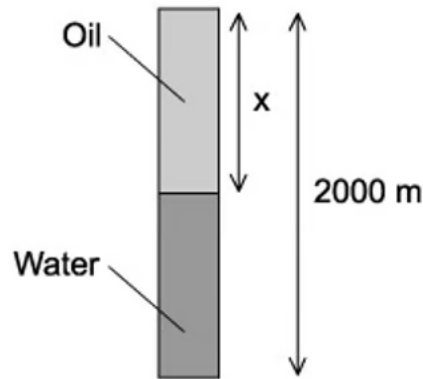
What is an estimate of the average pressure exerted on the lead by the α -particles ?

- A. $5.0 \times 10^{-15} \text{ Pa}$
- B. $5.0 \times 10^{-13} \text{ Pa}$
- C. $5.0 \times 10^{-11} \text{ Pa}$
- D. $5.0 \times 10^{-9} \text{ Pa}$

[1 mark]

Question 4

A bore-hole of depth 2000 m contains both oil and water as shown. The pressure due to the liquids at the bottom of the bore-hole is 17.5 MPa. The density of the oil is 830 kg m^{-3} and the density of the water is 1000 kg m^{-3}



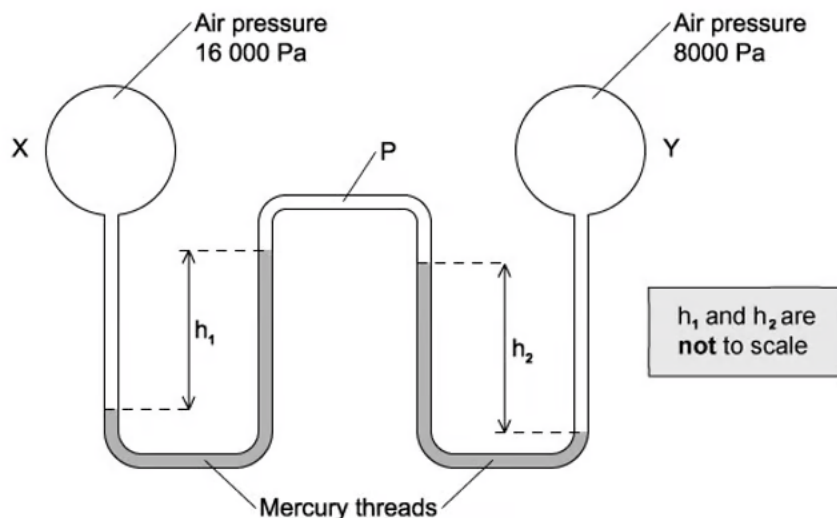
What is the depth x of the oil?

- A. 907 m
- B. 1000 m
- C. 1090 m
- D. 1270 m

[1 mark]

Question 5

Two bulbs X and Y containing air at different pressures are connected by a tube P which contains two mercury threads.



The density of mercury is $13\,600\text{ kg m}^{-3}$

Which pair of values of h_1 and h_2 is possible?

	h_1 / cm	h_2 / cm
A	4.0	2.0
B	6.0	6.0
C	12.0	18.0
D	18.0	12.0

[1 mark]

Question 6

1.5 m^3 of water is mixed with 0.50 m^3 of alcohol. The density of water is 1000 kg m^{-3} and the density of alcohol is 800 kg m^{-3}

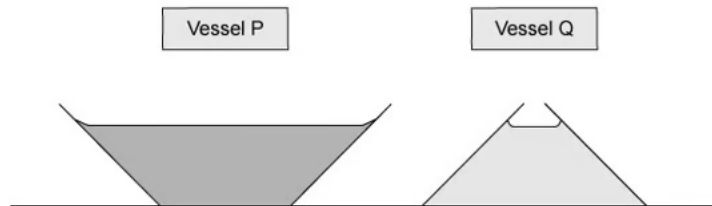
What is the density of the mixture with volume 2.0 m^3 ?

- A. 850 kg m^{-3}
- B. 900 kg m^{-3}
- C. 940 kg m^{-3}
- D. 950 kg m^{-3}

[1 mark]

Question 7

The diagram shows two vessels, P and Q, both with sides inclined at 45°



Vessel P tapers outwards and vessel Q tapers inwards, as shown.

Both vessels contain a liquid. The depth of the liquid in the vessels is the same. The liquid in vessel P is twice as dense as the liquid in vessel Q.

What is the ratio $\frac{\text{pressure due to the liquid on the base of P}}{\text{pressure due to the liquid on the base of Q}}$?

- A. $\frac{2}{1}$
- B. $\frac{\sqrt{2}}{1}$
- C. $\frac{1}{\sqrt{2}}$
- D. $\frac{1}{2}$

[1 mark]

Question 8

At a depth of 20 cm in a liquid of density 1800 kg m^{-3} , the pressure due to the liquid is p .

Another liquid has a density of 1200 kg m^{-3}

What is the pressure due to this liquid at a depth of 60 cm?

A. $\frac{p}{2}$

B. $\frac{3p}{2}$

C. $2p$

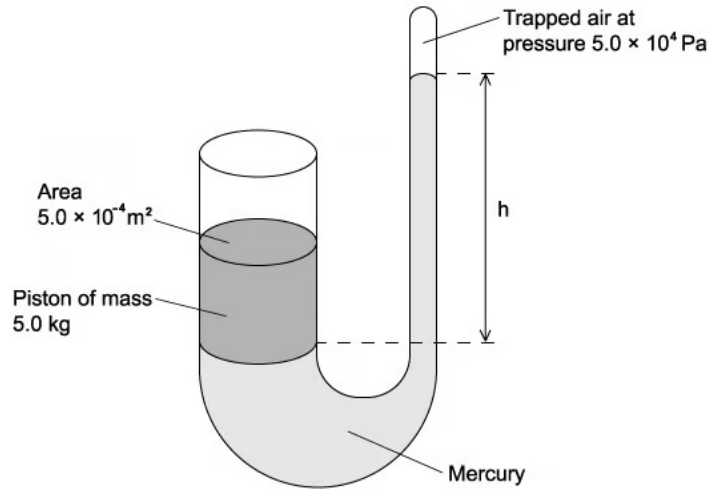
D. $3p$

[1 mark]

Question 9

A U-tube closed at one end contains mercury. Air at a pressure of $5.0 \times 10^4 \text{ Pa}$ is trapped at the closed end. The other end is open to the atmosphere and is fitted with a piston of mass 5.0 kg and cross-sectional area $5.0 \times 10^{-4} \text{ m}^2$

The density of mercury is $13\,600 \text{ kg m}^{-3}$ and atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$



What is the height of the mercury column?

- A. 37 cm
- B. 44 cm
- C. 74 cm
- D. 110 cm

[1 mark]

Question 10

A ball is falling at terminal speed in still air. The forces acting on the ball are upthrust, viscous drag and weight.

What is the order of increasing magnitude of these three forces?

- A. upthrust $\rightarrow$ viscous drag $\rightarrow$ weight
- B. viscous drag $\rightarrow$ upthrust $\rightarrow$ weight
- C. viscous drag $\rightarrow$ weight $\rightarrow$ upthrust
- D. weight $\rightarrow$ upthrust $\rightarrow$ viscous drag

[1 mark]



Model Answers: Medium

1

The correct answer is **A** because:

- Even though the cuboids have different densities, the question states that they are all **held** at the same depth, meaning an external force is keeping them from moving
- Pressure stays constant at the same depth beneath the water so all the cuboids have the same magnitude of upthrust
- Let's consider what would happen to the cuboids if the external force holding them in place was removed
- The equation of hydrostatic pressure is $P = \rho gh$
- Pressure is equal to $P = \frac{F}{A}$ so combining the two leads to:
 - $F = \rho ghA$
- We can see force is directly proportional to density, so the higher the density, the larger the upthrust on the object

B is incorrect because

while this is not true when the cuboids are held in place, the density of X is twice the density of Y. $F \propto \rho$ so when the external force is removed, the upthrust of the water on W is twice the upthrust of the water on Y.

C is incorrect because

while this is not true when the cuboids are held in place, this is also true when the external force is removed as the density of Y is twice that of X so the upthrust of the water on Z is half the upthrust of the water on X.

D is incorrect because

upthrust will always act on a body submerged in a liquid, this is due to the pressure exerted by the water molecules, so the upthrust on the cuboids will not be zero while it is submerged.

2

The correct answer is **A** because:

- Let the depth in the sea at which pressure is 150 kPa be height h
- The pressure at h is due to the amount of water above this point and due to the atmospheric pressure at sea level
- The equation of hydrostatic pressure is $P = \rho gh$
- Pressure at depth h = pressure due to height h of water + atmospheric pressure
 - $150\,000 = \rho gh + 100\,000$
 - $\rho gh = 50\,000$
 - $h = \frac{50\,000}{\rho g} = \frac{50\,000}{1020 \times 9.81} = 4.996 = 5.0\text{ m}$
- The pressure is 150 kPa at 5 m below sea level

3

The correct answer is **C** because:

- Pressure is equal to $P = \frac{F}{A}$
 - Number of particles per second = $\frac{N}{t} = 5.0 \times 10^4$
 - Area = $1.0\text{ cm}^2 = 1 \times 10^{-4}\text{ m}^2$
 - Mass of one particle = $6.6 \times 10^{-27}\text{ kg}$
 - Velocity = $1.5 \times 10^7\text{ m s}^{-1}$
- Force is defined as the rate of change of momentum $F = \frac{\Delta p}{\Delta t}$
- Force exerted due to one particle = $\frac{mv}{t}$
- Total force due to N particles = $N \left(\frac{mv}{t} \right) = \frac{N}{t} \times mv$
- Pressure

$$P = \frac{F}{A} = \frac{N}{t} \times \frac{mv}{A} = \frac{(5.0 \times 10^4) \times (6.6 \times 10^{-27}) \times (1.5 \times 10^7)}{1 \times 10^{-4}} = 4.95 \times 10^{-11}\text{ Pa}$$
- The pressure exerted by the α -particles is $5.0 \times 10^{-11}\text{ Pa}$

4

The correct answer is **D** because:

- The pressure at the bottom of the bore-hole is the sum of the pressures exerted by water and oil on the bottom which is equal to 17.5 MPa
- The equation of hydrostatic pressure is $P = \rho gh$
- For the column of water:
 - Depth of water, $h = 2000 - x$
 - Density of water, $\rho = 1000 \text{ kg m}^{-3}$
 - Acceleration due to gravity, $g = 9.81 \text{ m s}^{-2}$
- Pressure exerted by water = $(2000 - x)(1000)(9.81)$
- For the column of oil:
 - Depth of water, $h = x$
 - Density of water, $\rho = 830 \text{ kg m}^{-3}$
 - Pressure exerted by oil = $(830)(9.81)x$
- The sum of the pressures is 17.5 MPa
 - $(2000 - x)(1000)(9.81) + (830)(9.81)x = 17.5 \times 10^6$
 - $(2000 \times 1000 \times 9.81) - 17.5 \times 10^6 = 1000 gx - 830 gx$
 - $2.12 \times 10^6 = 170 gx$
 - $x = \frac{2.12 \times 10^6}{170 \times 9.81} = 1271 \text{ m}$
- The depth of oil in the bore-hole is about 1270 m

5

The correct answer is **D** because:

- The equation of hydrostatic pressure is $P = \rho gh$
- Let the pressure at P = p
- Let's consider the two sides separately
- On the left section:
 - $16\,000 - p = h_1 \rho g$ (1)
- On the right section:
 - $8000 - p = h_2 \rho g$ (2)
- If we take equation (2) from equation (1)
 - $16\,000 - 8000 = (h_1 - h_2) \rho g$
 - $h_1 - h_2 = \frac{8000}{\rho g} = \frac{8000}{13\,600 \times 9.81} = 0.06 \text{ m}$
- The difference between heights and is 0.06 m or 6 cm. The only set of values which matches this difference is choice D where $h_1 = 18.0$ and $h_2 = 12.0 \text{ cm}$

6

The correct answer is **D** because:

- Density of mixture = $\frac{\text{total mass}}{\text{total volume}}$
- Total volume = 2.0 m^3
- For water:
 - Density, $\rho = 1000 \text{ kg m}^{-3}$
 - Volume, $V = 1.5 \text{ m}^3$
 - Mass, $m = \rho V = 1000 \times 1.5 = 1500 \text{ kg}$
- For alcohol:
 - Density, $\rho = 800 \text{ kg m}^{-3}$
 - Volume, $V = 0.5 \text{ m}^3$
 - Mass, $m = \rho V = 800 \times 0.5 = 400 \text{ kg}$
- Total mass = $1500 + 400 = 1900 \text{ kg}$
- Density of mixture = $\frac{\text{total mass}}{\text{total volume}} = \frac{1900}{2} = 950 \text{ kg m}^{-3}$
- The density of the mixture is 950 kg m^{-3}

7

The correct answer is **A** because:

- The equation of hydrostatic pressure is $P = \rho gh$
- The pressure P depends only on the depth h , the density ρ of the liquid and the acceleration of free fall g . The volume of the container does not affect the pressure at any point
- Let the density of the liquid in vessel Q = ρ and the density of the liquid in vessel P = 2ρ
- The depth h of the liquids is the same
- For vessel P:
 - $P = h(2\rho)g = 2\rho gh$
- For vessel Q:
 - $P = \rho gh$
- Ratio: $\frac{\text{pressure due to the liquid on the base of P}}{\text{pressure due to the liquid on the base of Q}} = \frac{2\rho gh}{\rho gh} = \frac{2}{1}$

- Ratio: $\frac{\text{pressure due to the liquid on the base of } P}{\text{pressure due to the liquid on the base of } Q} = \frac{2\rho gh}{\rho gh} = \frac{2}{1}$

8

The correct answer is **C** because:

- The equation of hydrostatic pressure is $P = \rho gh$
- Pressure of liquid 1, P_1 :
 - Pressure, $P_1 = p$
 - Depth of water, $h = 20 \text{ cm} = 0.2 \text{ m}$
 - Density of water, $\rho = 1800 \text{ kg m}^{-3}$
 - Acceleration due to gravity, $g = 9.81 \text{ m s}^{-2}$
- $P_1 = \rho gh_1 = 1800 \times 9.81 \times 0.2$
- Pressure of liquid 2, P_2 :
 - Depth of water, $h = 60 \text{ cm} = 0.9 \text{ m}$
 - Density of water, $\rho = 1200 \text{ kg m}^{-3}$
- $P_2 = \rho_2 gh_2 = 1200 \times 9.81 \times 0.6$
- $\frac{P_1}{P_2} = \frac{\rho_1 gh_1}{\rho_2 gh_2} = \frac{1800 \times 9.81 \times 0.2}{1200 \times 9.81 \times 0.6} = \frac{360}{720} = \frac{1}{2}$
- $\frac{p}{P_2} = \frac{1}{2}$
- $P_2 = 2p$
- The pressure of liquid 2 at 60 cm is two times larger than the pressure of liquid 1 at 20 cm

9

The correct answer is **D** because:

- The equation of hydrostatic pressure is $P = \rho gh$
- At the same level in the liquid mercury, the pressure on both sides of the U-tube should be the same
- Consider the pressure at the lower level indicated – where the piston is in contact with the liquid mercury
- Pressure at the left end = Pressure at the right end
- The pressure on the left is due to the weight of the piston and due to

atmospheric pressure above the piston

- The pressure on the right is due to the height h of the column of mercury and due to the trapped air above the mercury
- The pressure of the trapped air plus the pressure of the mercury column of height h must equal total pressure supplied by the piston and the atmosphere
- Pressure due to piston = $\frac{\text{Force}}{\text{Area}} = \frac{\text{Weight}}{\text{Area}} = \frac{mg}{A}$
- Pressure at left end = Pressure at right end
- Pressure due to piston and atmosphere = Pressure due to mercury column and trapped air
 - $\frac{mg}{A} + P_{atm} = \rho gh + P_{air}$
 - $\frac{5.0 \times 9.81}{5.0 \times 10^{-4}} + 1.01 \times 10^5 = (h \times 13\,600 \times 9.81) + 5.0 \times 10^4$
 - $9.81 \times 10^4 + 1.01 \times 10^5 - 5.0 \times 10^4 = 133\,416\,h$
 - $149\,100 = 133\,416\,h$
 - $h = \frac{149\,100}{133\,416} = 1.1\,m$
- Therefore, the height of the mercury column is 110 cm

10

The correct answer is **A** because:

- At terminal speed, the resultant force on the ball is zero. As weight is the only downward force, it must be the largest of the three
- The upthrust is a result of the force due to the difference in pressure at the top and bottom of the ball
 - Pressure, $P = \rho gh$
 - Force, $F = PA$
- Air is not very dense (low ρ) and the difference in height h from the top of the ball to its bottom is also very small. Thus, the upthrust force is small
- Viscous drag (or air resistance), on the other hand, will almost be as large as the weight when a body is travelling at terminal velocity
- So, from smallest to largest force: upthrust $\rightarrow$ viscous drag $\rightarrow$ weight

It is a common misconception that the Archimedean upthrust on a body in air is quite large, however, this is incorrect, and it is the opposite which is true. The upthrust will only be about a thousandth or less than the weight of the solid